

Introduction to Data Analytics Data Science and Data Mining

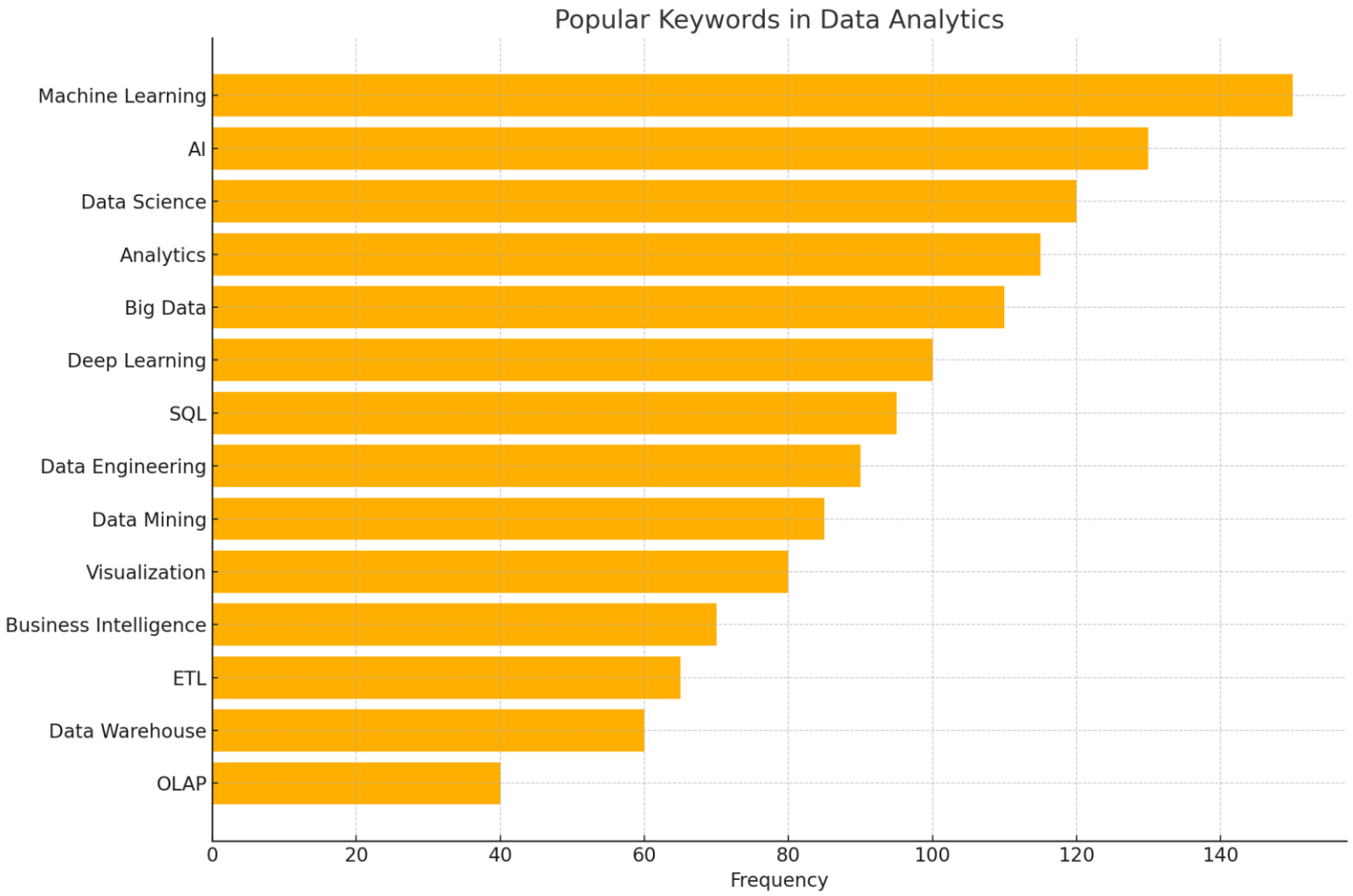
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Most popular keywords related to Data Analytics, Data Science and Data Mining



Why should you learn Data Analytics?

- Explore and prepare data
- Communicate insights
- Visualize data
- Build predictive models
- Work effectively with AI tools
- Ask meaningful business questions
- Support data-driven decisions

Analytics skills are becoming essential across all industries.

Why is analytics important now?

- Millions of searches are performed
- Thousands of online purchases occur
- Millions of videos are watched
- Massive amounts of sensor data are generated
- Billions of messages are exchanged
- …

Data alone has little value.
The value comes from transforming data into decisions.

Why is analytics important now?

- Lots of data is being collected and warehoused
  - Web data, e-commerce
  - purchases at grocery stores
  - Bank/Credit Card transactions
  - Social networks, Big data, …
  - And more Complex Data, …





- Computers have become cheaper and more powerful
- Competitive Pressure is Strong
- **True value is not the data itself but the interesting patterns, information and knowledge we can extract from those data.**

Data Analytics in Everyday Life

- Google Maps → Best route recommendation
- Netflix → Personalized recommendations
- TikTok/YouTube → Content ranking
- Banking Apps → Fraud detection
- Grab/Uber → ETA prediction and surge pricing
- ChatGPT → Conversational analytics and AI
- ETC.

These systems collect Data

From Data to Better Decisions

Analytics transforms raw data into value

- Raw data alone does not create value
- Data must become information
- Information must become insight
- Insight must drive action
- Action creates business value

- Data → Information → Knowledge → Action
- This class focus on the first three stages

=> The goal of analytics is not building models—it is making better decisions.

Why Organizations Invest in Analytics

- Make smarter decisions
- Reduce costs
- Increase revenue
- Improve customer experience
- Predict future events
- Automate decision-making

The goal is creating business value.

Data Analytics

- **Data Analytics** is the process of **examining and transforming data into insight** for making better decisions.

- **Data analytics** is the process of examining, transforming, and summarizing data with the goal of extracting meaningful insights, informing decisions, and uncovering patterns and knowledge

- **Goal of data Analytics** is to get actionable insights resulting in smarter decisions and better business outcomes.

- Techniques can be **data visualization**, **data warehousing**, **data mining**, **machine learning**, **data science, AI**, **Generative AI** etc.
- **Generative AI** (ChatGPT, Claude, Copilot) is transforming analytics

Data vs Information vs Knowledge

- Data:
  - Raw organized facts and observations
    - Examples: 07:45, 38°C, 120 THB, 4.7 stars
  - Data may be incomplete or noisy
  - Data alone has little meaning, and without context is difficult to interpret.
- Information:
  - Data that has been *processed* into a form that is meaningful to the recipient (context)
  - Information = Data + Context
  - Context transforms data into information.
  - Examples
    - Average travel time is 35 minutes
    - Coffee sales increase by 30% on rainy days
    - Information answers who, what, when, and where

Data Analytics ในชีวิตประจำวัน

Smarter decisions => Faster actions

- **กิจกรรมที่เลือก**: การเดินทางไปมหาวิทยาลัยด้วย Grab/BTS
- **ข้อมูล (Data) ที่เกี่ยวข้อง**:
  - เวลาที่ออกจากบ้าน (7:30, 8:00, 8:15 น.)
  - ระยะทางที่เดินทาง (12.5 กม.)
  - ค่าใช้จ่าย (Grab 80-120 บาท, BTS 42 บาท)
  - เวลาที่ใช้ในการเดินทาง (25-45 นาที)
  - สภาพการจราจร (ติด/ไม่ติด)
  - สภาพอากาศ (ฝนตก/แดดจัด/ปกติ)
- **ข้อมูลสารสนเทศ (Information):**
  - **ช่วงเวลา** 7:30-8:00 **น.** ใช้เวลาเดินทางเฉลี่ย 28 นาที ค่าใช้จ่าย 85 บาท
  - **ช่วงเวลา** 8:00-8:30 **น.** ใช้เวลาเดินทางเฉลี่ย 40 นาที ค่าใช้จ่าย 110 บาท
  - **วันฝนตก** ค่า Grab เพิ่มขึ้น 30-40% และเวลารอนานขึ้น
- **BTS** เวลาคงที่ 25 นาที แต่ต้องเดินเพิ่ม 10 นาที
  - **ความรู้ (Knowledge):**
  - **ข้อสรุป**: ออกจากบ้านก่อน 8:00 น. จะประหยัดเวลาและเงิน ในวันฝนตกควรใช้ BTS เพราะเวลาแน่นอนและประหยัดกว่า

Data-driven thinking can lead to better outcomes in even simple scenarios

Data vs Information vs Knowledge

- Knowledge = Combination of ideas, rules, and procedures that guide actions and decisions
  - Knowledge = Information + Experience
  - Knowledge helps answer:
    - What should we do?
  - Examples
    - Leave home before 7:45 AM to avoid traffic
    - Use BTS instead of Grab when it rains



Analytics Everyday Example: Going to University

- Data: departure time, travel time, weather
- Information: average travel time by time period
- Knowledge: leave before 8:00 AM
- Action: check weather and choose transportation

CoffeeConnect Case: Data, Information, Knowledge

Data
(Raw Observations)

- Time of purchase: 7:30, 9:00, 13:00
- Menu items purchased: Iced Latte, Croissant, Americano
- Customer segment: Office Worker, Student
- Weather condition: Sunny, Rainy
- Use of promotions: Yes / No
- Satisfaction rating: 1–5 stars

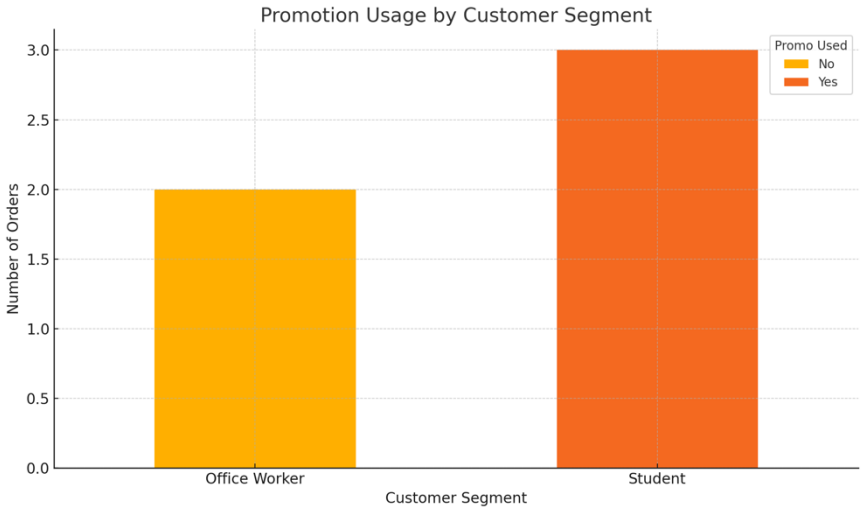
Information
(Processed & Organized)

- Most iced lattes are sold between 7:30–9:30 AM
- 70% of students use "Buy 1 Get 1" promotions
- Customers buying icellattes tend to also buy a snack
- Satisfaction ratings are lower during rainy mornings for Grab orders

Knowledge
(Actionable Insight)

- Promote snack bundles with iced lattes during morning hours
- Target students with special discounts in the afternoon
- Recommend using BTS delivery channels during rainy weather
- Send 20% discount coupons to customers who haven't visited in 2 weeks

Time	Item Purchased	Segment	Weather	Used Promo	Rating
07:30	Iced Latte, Croissant	Student	Sunny	Yes	5
09:00	Iced Latte	Office Worker	Rainy	No	3
13:00	Americano, Croissant	Student	Sunny	Yes	4
08:45	Iced Latte	Student	Rainy	Yes	2
13:00	Iced Latte, Americano	Office Worker	Rainy	No	4



Check Your Understanding

- "92" → Data
- "Average exam score is 92" → Information
- "Students attending class score above 90" → Knowledge
- "Add weekly quizzes to increase attendance" → Action

CoffeeConnect Case

- Data: purchase time, weather, menu items
- Information: students use promotions more
- Knowledge: target afternoon promotions to students
- Action: launch personalized campaigns

⇒ Analytics creates business value.
⇒ Analytics creates value by turning data into action.

The Analytics Pipeline

- Data → Information → Knowledge → Action → Value

- Transactions → Customer insights → Promotions
- The goal is not collecting data
- The goal is creating value => Better outcomes

Data-driven decision can lead to better outcomes in even simple scenarios

CoffeeConnect Case

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(Raw Observations)

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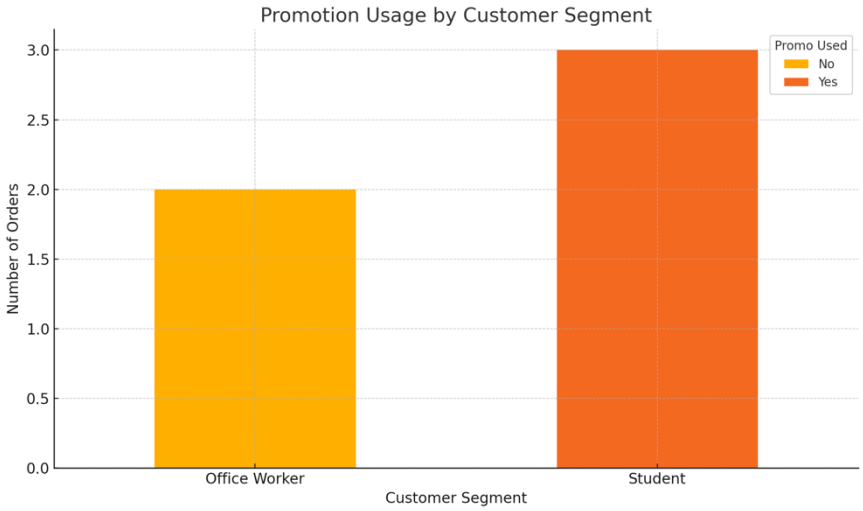
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Benefits of Data Analytics

- **Smarter decisions => Faster actions**

- **Resource and Time Reduction**
  - Prediction of product sale => Reduce cost of inventory (Inventory optimization)
  - Google Map best route suggestion => Reduce searching time
  - Fraud prevention
  - Etc.

- **Value Adding**
  - Product recommendation => Increase sale figures
  - Early cancer prediction => Treatment preparation
  - Trend detection
  - Etc.

- **Innovation Creation**
  - New products and inventions
  - Self-driving cars
  - AI-powered services (e.g., Uber ETA, Spotify mood detection) => Enhances **user experience**
  - Etc.

Various applications of data analytics

- Weather prediction
- Medical diagnosis
- Financial markets
- Resource management
- Smart buildings and cities
- Etc.

2023 Senior Project Examples

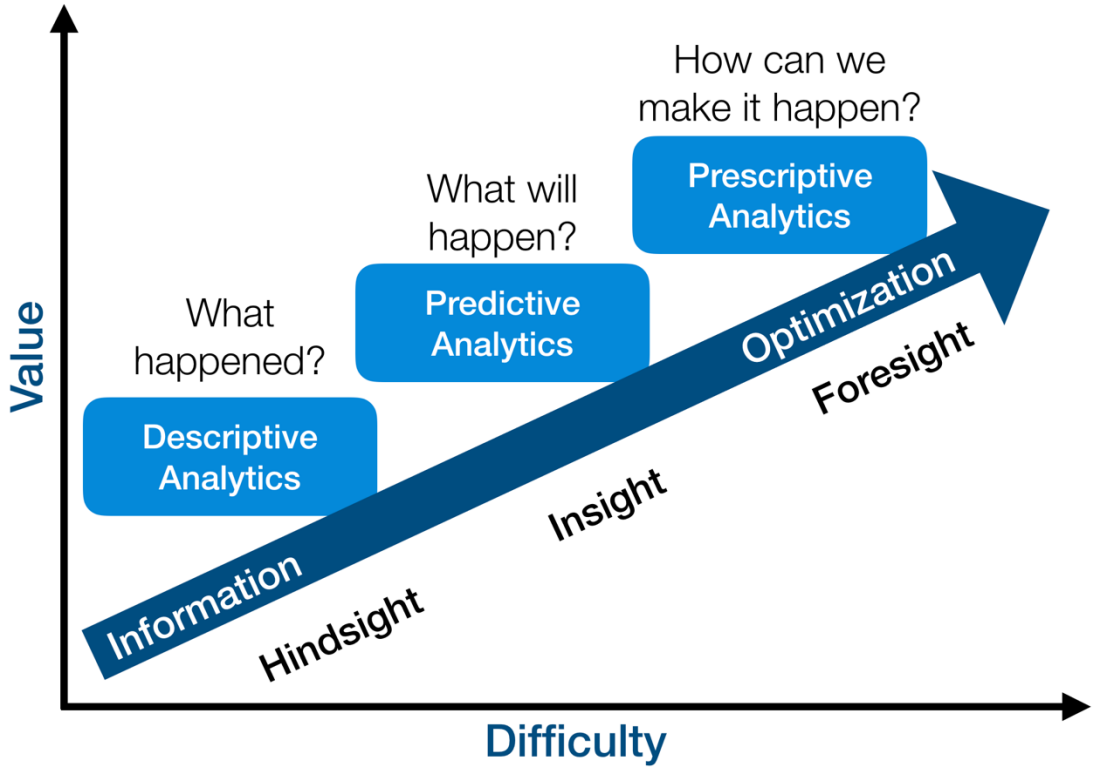




Data Analytics Types

Descriptive → What happened?
Diagnostic → Why did it happen?
Predictive → What will happen?
Prescriptive → What should we do?

Each level provides greater business value.



Descriptive Analytics

- Summarizes historical and current data
- Uses dashboards, reports, and visualizations
- Answers: What happened?
- Examples: sales reports, website traffic, KPIs

Descriptive Analytics (การวิเคราะห์เชิงพรรณนา)

- **Descriptive analytics** looks at data and analyze past and current events for insight as to how to approach future events.
- It looks at the past performance and understands the performance by analyzing historical data to understand the cause of success or failure in the past.
- Analyze to find some conclusions.

- Some **descriptive techniques**:
  - Data Warehouse, Data Queries, Reports
  - Data dashboard and visualization techniques
  - Descriptive Statistics,
  - etc.

Application Domains of Descriptive Analytics

- **Sale**
  - Average sales per quarter for the 3 years ago
  - Characteristics of highest profit customers of the last three years
  - For each month, highest sales by customer characteristics
  - What percentage of customers who visit your website click and buy?
- **Health**
  - Which month has the highest % of people sick with influenza?
  - Age range having the most deaths from COVID?
- **Education**
  - Students from which school have achieved the highest percentage of grades in the past 5 years?
- **Agriculture**
  - For each quarter of the last year, show export volume of jasmine rice
- **Telecom**
  - During Songkran, what types of passengers like to travel to Chiang Mai?
- **Etc.**

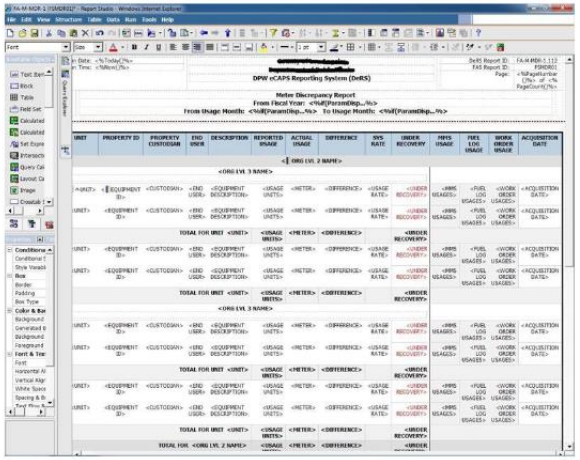
DASHBOARDS AND SCORECARDS

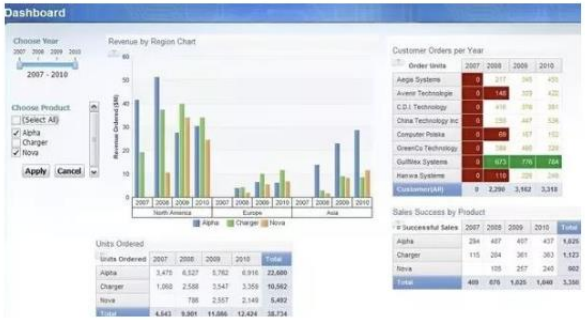


Results of Descriptive Analytics Tasks

Traditional BI

- Ability to combine data from multiple sources (vs Target, vs Budget, comparison)
- Great Reporting Capabilities (Cube, OLAP, Metadata)
- Need data integration
- Rich Reporting features (filtering / exporting / formatting)
- Some graphical element added
- Allow users to ad-hoc query





Diagnostic Analytics

- Investigates the root causes of events
- Uses drill-down, segmentation, and correlation analysis
- Answers: Why did it happen?
- Examples: Why did sales decrease? Why did customers churn?

Diagnostic analytics explains the reasons behind patterns.

Predictive Analytics

- Forecasts future outcomes using historical data
- Uses machine learning and statistical models
- Answers: What will happen?
- Examples: demand forecasting, fraud detection, customer churn

Predictive analytics estimates probabilities—not certainties.

Predictive Analytics (การวิเคราะห์เชิงพยากรณ์)

- Use data to determine the probable outcome of an event.
- Turn the data into valuable, actionable information.
- Holds a variety of statistical techniques from modeling, machine, learning, data mining, and game theory that analyze current and historical facts to make predictions about a future event
- Some **predictive techniques**:
  - Linear Regression
  - Time series analysis and forecasting
  - Data Mining, etc.

Application Domains of Predictive Analytics

- **Sale**
  - What will the next month's sales be?
  - Next month, which customers will leave to competitor?
  - Will this customer have the opportunity to come back to the store or not?
  - Based on this reviews, would the customer like the product or not?
- **Health**
  - How many COVID-19 cases will there be next month?
- **Education**
  - What will this student's grade point average be next semester?
- **Agriculture**
  - How many tons of jasmine rice will be exported next month?
- **Telecom**
  - During next year Songkran, how much will the death toll be?
- **Etc.**

Prescriptive Analytics

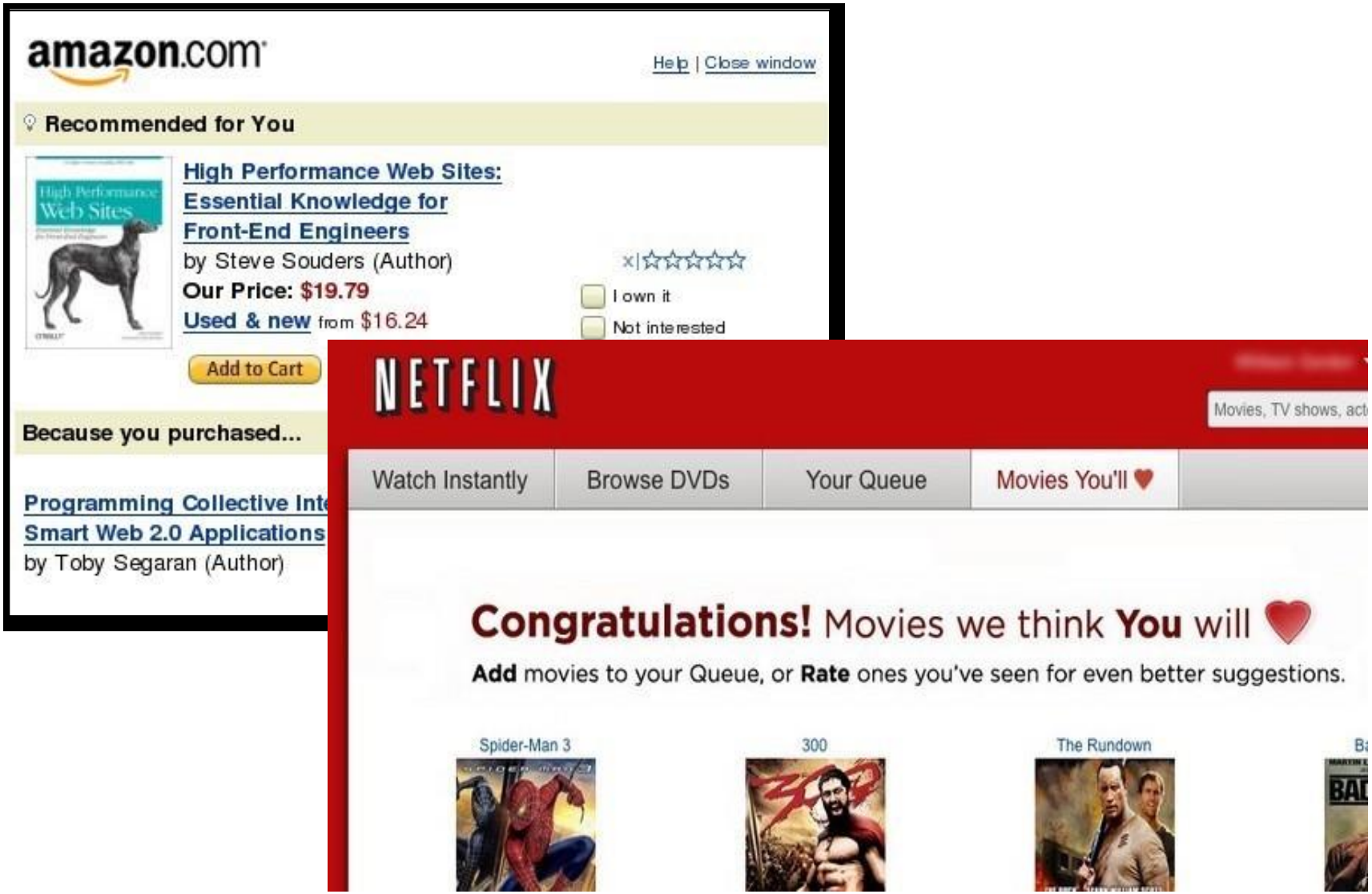
- Recommends actions based on predictions
- Uses optimization, recommendation systems, and simulation
- Answers: What should we do?
- Examples: route optimization, inventory planning, promotions

Prescriptive analytics supports better decisions.

Prescriptive Analytics (การวิเคราะห์เชิงแนะนำ)

- **Prescriptive Analytics** automatically synthesize data to make a prediction and then suggests a decision option to take advantage of the prediction.
- **Prescriptive Analytics** able to recommend for decision making (แนะนำเพื่อประกอบการตัดสินใจ)
- **Prescriptive analytics** goes beyond predicting future outcomes by also suggesting action benefit from the predictions and showing the decision maker the implication of each decision option.
- Further, **Prescriptive Analytics** can suggest decision options on how to take advantage of a future opportunity or mitigate a future risk and illustrate the implication of each decision option.
- Some **prescriptive techniques**:
  - Recommendation techniques
  - Collaborative filtering
  - Machine learning such as clustering
  - Etc.

Product Dev. From Prescriptive Analytics



Application Domains of Prescriptive Analytics

- **Sale**
  - Which products should I stock up next month?
  - Recommend interesting promotions
  - Please suggest products that should be placed on this page.
- **Health**
  - Symptoms like this, What are possible diseases ?
- **Education**
  - At which school should the department promote the new SKE-65 curriculum ?
- **Agriculture**
  - What should be planted in this coordinate?
- **Telecom**
  - Please suggest interesting places to visit for me.
- **Etc.**

One business problem often requires various analytics types.

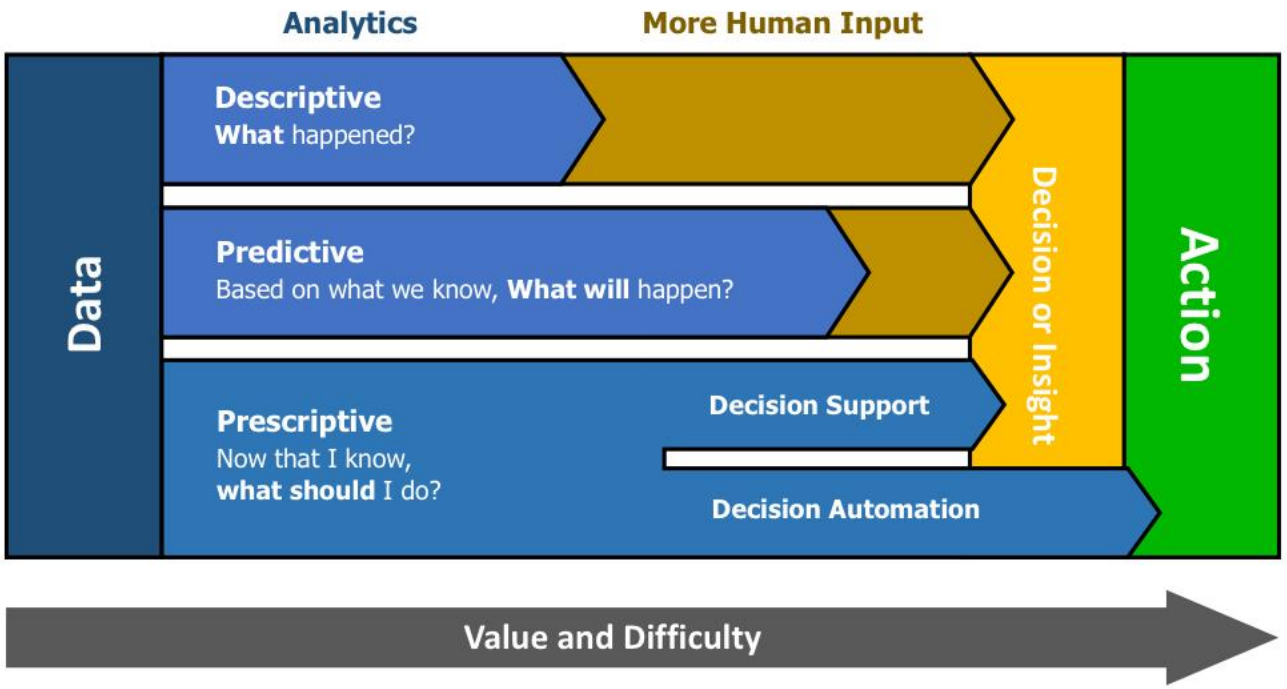
Online Retail Business (ธุรกิจค้าปลีกออนไลน์)

- **Descriptive analytics**
  - Summarizes and analyzes sales data, such as monthly sales figures, the number of new customers, and best-selling products.
  - Analyzes website traffic to provide an overall view of the business's current operational status.
- **Predictive analytics**
  - Forecasts future sales.
  - Predicts customer purchasing trends across different seasons.
  - Estimates website traffic during promotional periods.
  - Forecasts changes in the market and shifts in customer purchasing behavior.
- **Prescriptive analytics**
  - Recommends strategies to increase sales and profitability, such as designing effective marketing plans and optimizing inventory management.
  - Proposes special promotions or personalized offers for loyal customers.

Company Stock Analysis (วิเคราะห์หุ้นของบริษัท)

- **Descriptive analytics**
  - How the stock price increased/decreased over the past quarter?
  - In which month did this company's stock experience the highest increase or decrease?
- **Predictive analytics**
  - What will the price of this company's stock be in the next quarter?
  - Will this company's stock rise or fall in the next quarter if the Thai baht exchange rate increases?
- **Prescriptive analytics**
  - Which stock should be purchased to generate profit in the next quarter?
  - Which stock in the current portfolio should be sold now to maximize profit?

Types of Analytics



- Data supports decision-making and action
- However, data needs to undergo analytics before making decision

Main Steps of Analytics

1. Business Understanding
2. Data Acquisition & Storage
3. **Pre Preparation & Data Exploration Analysis (EDA)**
4. Modeling & Evaluation
   - **Data Mining**: Looking for patterns in data
   - **Data Science**
     - **Machine Learning**:
       Using data to build models and make predictions
     - **Deep Learning, LLM, …**
5. Deployment (e.g., API, dashboard)
6. Monitoring and Feedback

Step 1: Business Understanding

- Define the business problem
- Identify stakeholders and objectives
- Define success metrics
- Translate business goals into analytics questions

Start with the problem—not the data.

Step 2: Data Acquisition

- Identify relevant data sources
- Collect structured and unstructured data
- Integrate data from multiple systems
- Ensure data quality

Good analytics starts with good data.

Step 3: Data Preparation and EDA

- Clean missing and inconsistent data
- Transform and engineer features
- Explore patterns and relationships
- Visualize data distributions

Data preparation often takes 60–80% of project effort.

Step 4: Modeling and Evaluation

- Select appropriate algorithms
- Train and validate models
- Evaluate performance metrics
- Compare alternative approaches

The best model balances performance and interpretability.

Step 5: Deployment

- Deploy models through APIs or dashboards
- Integrate analytics into business processes
- Automate decision support
- Enable users to consume insights

Insights create value only when used.

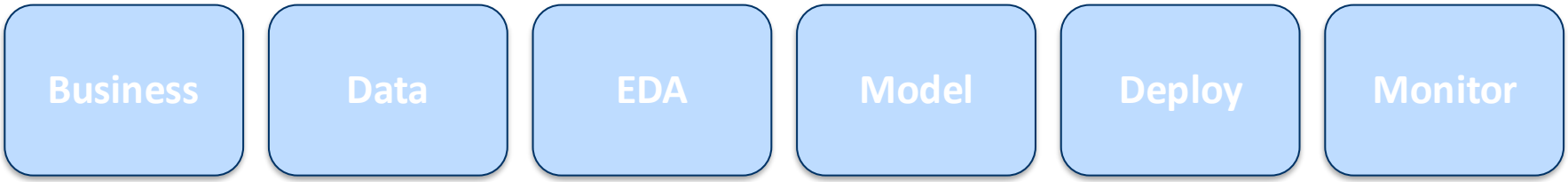
Step 6: Monitoring and Feedback

- Monitor model performance
- Detect data drift and concept drift
- Collect user feedback
- Retrain and improve models continuously

Analytics is a continuous improvement cycle.

Analytics Lifecycle

- Analytics projects are iterative, not linear
- Success depends on business understanding
- Models alone do not create value
- Deployment and monitoring are essential



When Not to Use Analytics

- Ethical risks: biased data, lack of consent
- When domain understanding is insufficient
- When data is poor quality or insufficient

- **Rule**: Garbage in = garbage out

The Big Picture: disciplines overlap but are not identical

- Analytics transforms data into decisions

- Artificial Intelligence enables intelligent systems
- Machine Learning learns patterns from data
- Deep Learning uses neural networks
- Data Science combines data, computation, and domain expertise

- Data Mining discovers hidden patterns in data
  - Includes classification, clustering, and association rules
  - Combines statistics, databases, and machine learning
  - Is a core component of Data Science

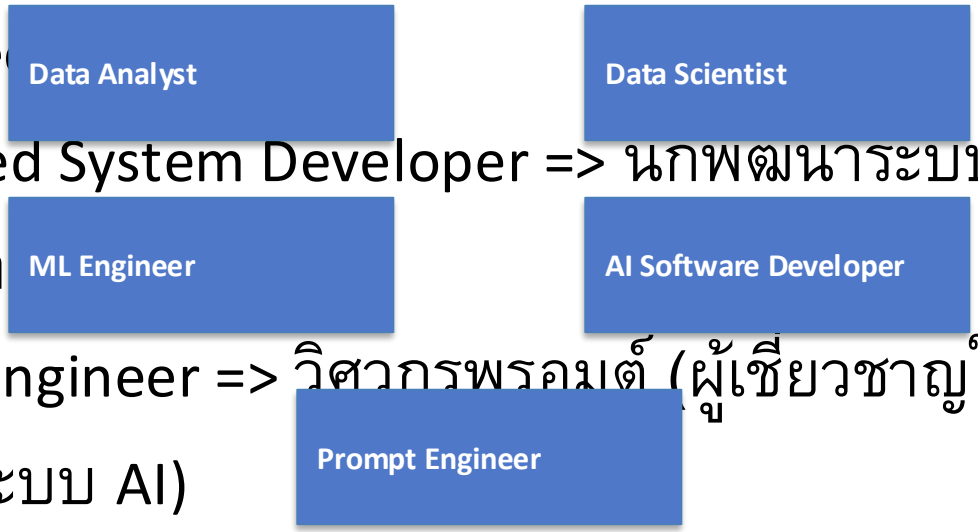
Comparing Key Disciplines

- Analytics → Supports decisions
- Data Mining → Discovers patterns
- Machine Learning → Builds predictive models
- Data Science → Solves problems using data
- Artificial Intelligence → Automates intelligent behavior

Different goals require different techniques.

Analytics Team

- Business Data Analyst => นักวิเคราะห์ข้อมูลธุรกิจ
- Data Engineer => นักวิศวกรรมข้อมูล
- Data Scientist => นักวิทยาศาสตร์ข้อมูล
- Machine Learning Specialist => ผู้เชี่ยวชาญด้านการเรียนรู้ของเครื่อง
- Software Developer => นักพัฒนาซอฟต์แวร์
- AI Engine
- AI-enabled System Developer => นักพัฒนาระบบที่มี AI เป็นส่วนประก
- Prompt Engineer => วิศวกรพรอมต์ (ผู้เชี่ยวชาญในการออกแบบคำสั่งให้ระบบ AI)





Business Data Analyst

- A **Business Data Analyst** is someone who possesses a deep understanding of a specific business domain.
- **Business** refers to specialized fields such as finance, agriculture, education, transportation, healthcare, etc.
- **Business Data Analyst** must also have a thorough understanding of how to utilize data to enhance the business's capabilities.
- Their main responsibilities include setting up analytical problems that can lead to value adding and creation for the business.
- This can involve reducing costs, increasing profits, or developing new products, among other strategies.

Data Scientist

- **Data scientists** come with a solid foundation of computing Techniques (programming, …), modeling(**AI**, **Data mining, ML**, etc.), statistics and math.
- A **Data Scientist** is a professional who understands data from a **business point of view**.
- Efficient in **picking the right problems**, which will add value to the organization after resolving it.
  - In charge of making predictions to help businesses take accurate decisions.
- Brilliance in business coupled with great communication skills, to deal with both business and IT leaders.

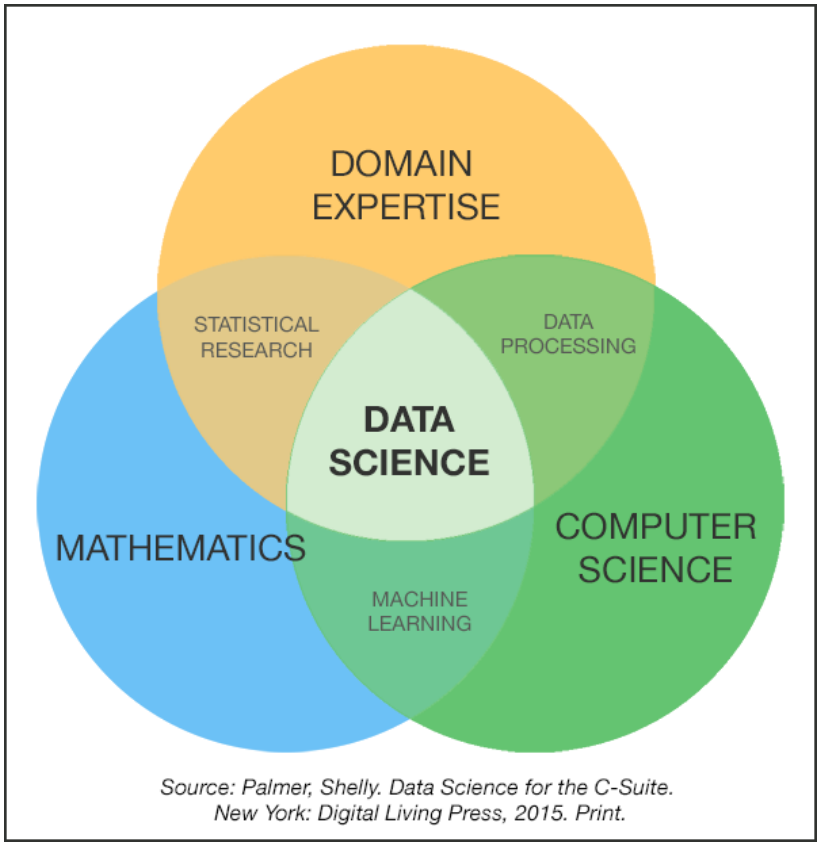




Data Scientist

"A data scientist is someone who is better at statistics than any software engineer and better at software engineering than any statistician."

© Someone at Quora



Qualification Required for Data Scientists

- Solid foundation of computing techniques, basic statistics and maths
- Familiar with data storing, retrieving, pre-processing, and visualization techniques:
  - SQL, No-SQL, etc. and Complex Data
  - Data Warehousing, ETL,
  - Exploration & Visualization, Etc.
- Master Modelling techniques such as Data mining, AI, Machine Learning, Deep- learning, etc.
- Familiar with Python programming (libraries, …) etc.
- Strong Business & Communication skills
- Strong understanding of Hadoop-based analytics such as MapReduce, Spark, Hive, etc.

Some data science Basic skills ….

- Basics => Maths, Stats, Computing Techniques, Python
- Data Science Process/Life-Cycle
- Data Exploration and Visualization
- Data-preparation & Pre-processing
  - Data collection & storage of various and new types of data
    - SQL/NoSQL
    - Open data, IoT data, Sensor data, etc.
  - Data manipulation/cleaning/pre-processing
  - Basic Modelling Techniques
    - Regression, Decision-Tree, etc.
    - NLP and ML and AI
  - Teamwork
  - Business & Communication Skills



More about Data Science

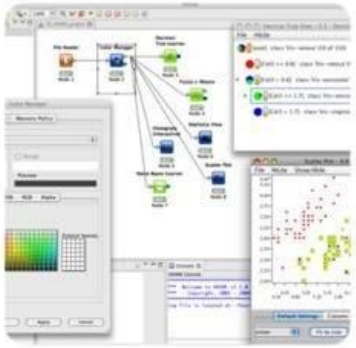
- **Multidisciplinary** field that combines various techniques, algorithms, and systems to extract insights and knowledge from **structured** and **unstructured** data.

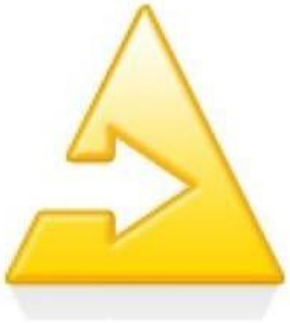
- **Multidisciplinary field** that combines:
  - Programming (Python, SQL, Spark)
  - Math & Statistics
  - Domain Expertise
  - Communication
  - Ethical Reasoning

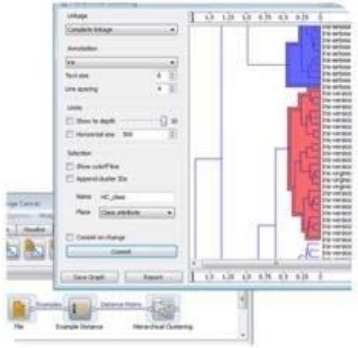
- The primary goal of data science is to use data to solve complex problems, make decisions, and drive innovation.

Data Analytics tools









DATA SCIENCE



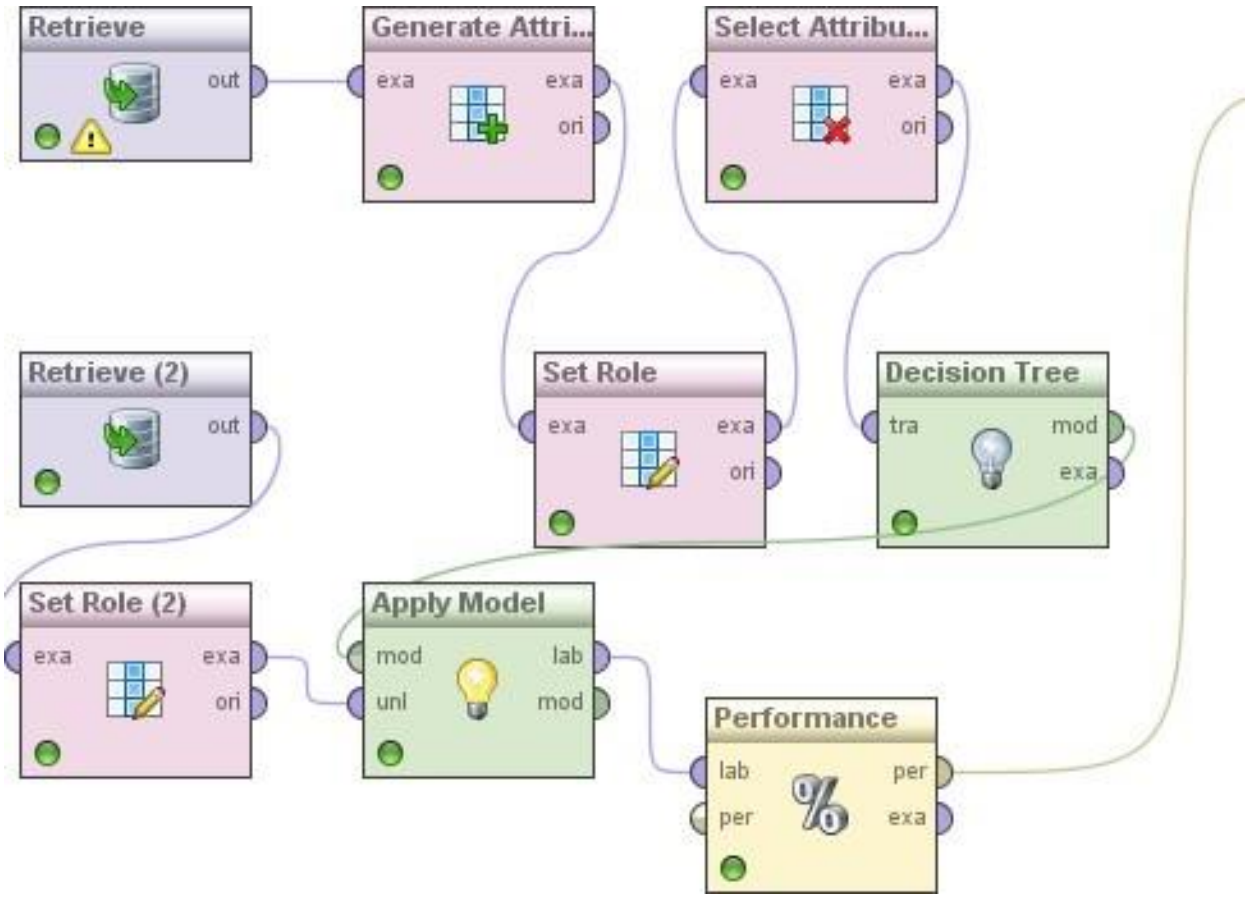




Example of Data Analytics Softwares

- SAS Enterprise Miner
- SQL Server
- R (Free)
- Weka (Free)
- RapidMiner (Free)
- Python analytic (Free)

Rapid Miner Screen



Python Classification Modelling Template

```
# Classification template

# Importing the libraries
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

# Importing the dataset
dataset = pd.read_csv('Social_Network_Ads.csv')
X = dataset.iloc[:, [2, 3]].values
y = dataset.iloc[:, 4].values

# Splitting the dataset into the Training set and Test set
from sklearn.cross_validation import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.25, random_state = 0)

# Feature Scaling
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)

# Fitting classifier to the Training set
# Create your classifier here

# Predicting the Test set results
y_pred = classifier.predict(X_test)

# Making the Confusion Matrix
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)
```

K-NN with Python

```python
# K-Nearest Neighbors (K-NN)

# Importing the libraries
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

# Importing the dataset
dataset = pd.read_csv('Social_Network_Ads.csv')
X = dataset.iloc[:, [2, 3]].values
y = dataset.iloc[:, 4].values

# Splitting the dataset into the Training set and Test set
from sklearn.cross_validation import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.25, random_state = 0)

# Feature Scaling
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)

# Fitting K-NN to the Training set
from sklearn.neighbors import KNeighborsClassifier
classifier = KNeighborsClassifier(n_neighbors = 5, metric = 'minkowski', p = 2)
classifier.fit(X_train, y_train)

# Predicting the Test set results
y_pred = classifier.predict(X_test)

# Making the Confusion Matrix
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)
```